

Project Requirements & Use Cases

Project Title

DeFi Staking Platform

Project Overview

The DeFi Staking Platform is a decentralized application (DApp) that allows users to stake cryptocurrency assets, view their staking balance, earn rewards, and withdraw funds securely through blockchain smart contracts.

The platform uses Solidity smart contracts for staking operations and a React.js frontend integrated with MetaMask and Ethers.js.

Objectives

- Provide a decentralized staking mechanism.
- Allow users to connect their wallets securely.
- Enable staking and withdrawal of funds.
- Display staking balances and rewards.
- Demonstrate Web3 and blockchain integration.

Functional Requirements

Wallet Connection

- Users should be able to connect MetaMask.
- The connected wallet address should be displayed.

Staking

- Users can stake ETH into the staking pool.
- The smart contract should store user balances.

Balance Tracking

- Users can view their staked balance at any time.

Reward Calculation

- The platform should calculate rewards based on the staked amount.
- Rewards are currently calculated as 10% of the user's stake.

Withdrawal

- Users should be able to withdraw their staked funds.

Smart Contract Integration

- The frontend should communicate with the deployed smart contract using Ethers.js.

Non-Functional Requirements

- Secure wallet authentication.

Project Requirements & Use Cases

- User-friendly interface.
- Fast transaction processing.
- Responsive design.
- Blockchain transparency.

Use Cases

Use Case 1: Connect Wallet

Actor: User

Steps:

1. User opens the application.
2. User clicks "Connect Wallet".
3. MetaMask requests permission.
4. Wallet address is displayed.

Expected Result:

Wallet successfully connected.

Use Case 2: Stake ETH

Actor: User

Steps:

1. User enters staking amount.
2. User clicks "Stake".
3. MetaMask confirms transaction.
4. Smart contract records stake.

Expected Result:

Balance increases successfully.

Use Case 3: View Balance

Actor: User

Steps:

1. User clicks "Check Balance".
2. Smart contract returns staked amount.

Expected Result:

Current staked balance is displayed.

Use Case 4: View Rewards

Actor: User

Steps:

Project Requirements & Use Cases

1. User clicks "Check Reward".
2. Smart contract calculates reward.

Expected Result:

Reward amount is displayed.

Use Case 5: Withdraw Funds

Actor: User

Steps:

1. User enters withdrawal amount.
2. User clicks "Withdraw".
3. Smart contract verifies balance.
4. Funds are transferred.

Expected Result:

Withdrawal completed successfully.

Similar Existing Protocols Studied

The following DeFi staking platforms were studied for understanding staking concepts:

- Lido Finance
- Rocket Pool
- Aave
- Compound Finance
- Ethereum Staking Protocols

These platforms inspired the staking, reward, and withdrawal mechanisms used in this project.

Conclusion

The DeFi Staking Platform demonstrates the core concepts of decentralized finance by allowing users to stake funds, track balances, calculate rewards, and withdraw assets through blockchain smart contracts. The project showcases Web3 integration using Solidity, React.js, MetaMask, and Ethers.js.